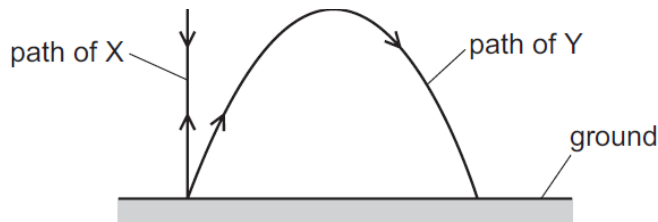


- 3 Two projectiles, X and Y, are launched into the air from the same place on level ground. They reach the same maximum height as shown.



Projectile X is launched vertically upwards and projectile Y is launched at an angle to the horizontal. Air resistance is negligible.

Which statement is correct?

- A X and Y are at rest at their maximum heights.
- B X and Y are launched with the same speed.
- C X and Y take the same time to return to the ground.
- D X and Y travel the same distance.

Ans: C

X and Y subjected to the same acceleration due to gravity in the vertical direction, so both reach the ground at the same time. Since X and Y reach the same maximum height, they must be launched with the same initial *vertical* speed (and NOT same speed since Y's initial vertical speed is  $u\sin\theta$ )

4. A tractor of mass  $1000\text{ kg}$  pulls a trailer of mass  $1000\text{ kg}$  via a tow-bar. The total resistance to